

Khaja Moinuddin Shaik Mohammed

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AI Engineer | 0→1 AI Products · Agentic Systems · Full-Stack AI

Hyderabad, India

SUMMARY

AI Engineer building 0→1 AI products and production AI systems. Experienced in agentic architectures, AI infrastructure, and solving reliability problems where off-the-shelf LLM behavior is insufficient.

SELECTED PROJECTS

- Decode** — Agentic Knowledge-to-Video System | *Python, FastAPI, React, Remotion, Redis, PostgreSQL* 2026
- Built a **multi-agent educational video generation product** that turns natural-language questions into explainable videos through specialized planning, teaching, scripting, motion-design, voice, and rendering agents coordinated by a central orchestrator.
 - Solved LLM-generated visual reliability problems by building a **custom visual component API** with coordinate transforms, collision detection, boundary detection, text bounding boxes, and word-level timing.
 - Implemented **visual validation, self-correction, DAG-based artifact lineage, and content-hash caching** to repair invalid scenes and incrementally regenerate only affected work.
- Attest** — Agentic RAG & Verification System | *Python, LangGraph, FastAPI, PostgreSQL, Redis* 2026
- Built an end-to-end grounded question-answering system that retrieves evidence, generates answers, verifies individual claims, and performs bounded revision before escalating uncertain responses.
 - Reduced failing citations from **9.87% to 4.15%** through claim-level verification using an independent judge model; implemented hybrid retrieval with **pgvector, PostgreSQL full-text search, reranking, and Redis-cached LLM calls**.
- Smart Turn** — Audio Turn Detection for Voice Agents | *PyTorch, Whisper, Transformers, ONNX* 2026
- Adapted the **Whisper-tiny encoder** with attention pooling and a classification head for lightweight speech turn-completion detection, achieving **93.9% accuracy on Hindi and 93.7% on English**.
 - Optimized an **8-second context** model and exported a single ONNX graph, achieving **~38 ms/clip on CPU**.

EXPERIENCE

- Applied Scientist Intern — Amazon** Jan 2026 – Jun 2026
RBS Sciences *Bangalore, India*
- Built a hybrid **ML/LLM pipeline** that organized millions of customer feedback records into hierarchical issue taxonomies with **0.74 F1**, outperforming the manual scientist baseline (**0.71 F1**).
 - Built an explainable **LLM evaluation framework** adopted by **12 Amazon business teams** to evaluate generated taxonomies.
 - Reduced new-business onboarding from **5–7 days to under 18 hours** through self-calibrating extraction that eliminated per-domain prompt engineering.
- AI Intern — Intel Corporation** May 2025 – Jul 2025
Intel Unnati Industrial Training Program *Hyderabad, India*
- Reduced model size by **30%** and inference latency by **35%** on GPU-free edge devices using knowledge distillation and quantization in PyTorch.

RESEARCH

Universal Anecdote Miner (UAM) — First Author, AMLC 2026 submission
LUMEN — Third Author, EMNLP 2026 submission

EDUCATION

- Mahatma Gandhi Institute of Technology** Oct 2022 – Jul 2026
B.Tech, Computer Science (AI and Machine Learning) *Hyderabad, India*
- GPA 9.09/10.0 · Amazon ML Summer School 2025** — 3,000 selected from 165,000+ applicants (~top 2%).

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, C++, Java
AI: LLMs, RAG, AI Agents, Multi-Agent Orchestration, LangGraph, PyTorch, Transformers, NLP, Speech
Product & Backend: FastAPI, React, PostgreSQL, Redis, REST APIs, SSE, Async Python, Docker
Infrastructure: AWS (Bedrock, SageMaker, S3, EC2), GCP, Git, Langfuse, ONNX